

LESSON 135 (0.5) GROUND (2.0) SOLO

Lesson Objective

The purpose of this lesson is to continue to enhance the applicant's confidence and abilities to operate the aircraft solo. During this lesson the applicant will practice normal and crosswind takeoffs and landings, slow flight, stalls, steep turns, and ground reference maneuvers.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 14 - Airport Operations.

Instructor Guidance

- Verify the student's solo endorsement currency before every unsupervised solo flight. Per 14 CFR §61.87, the endorsement is valid for 90 calendar days. If expired or approaching expiration, re-endorse before the student departs.
- Before releasing the student, quiz them on the unsupervised solo plan of action questions listed in the lesson — expected runway, wind direction and correction technique, anticipated ATC departure instructions, and go-around trigger scenarios. If the student cannot answer these confidently, delay the solo and address the gaps.
- Track the student's solo flight using FlightRadar24 or FlightAware for the duration of the flight. Confirm that the student is flying in the correct practice area and completing the required line items rather than remaining in the pattern.
- Remind the student before departure that all landings must be full-stop taxi-backs — no touch-and-go landings. This is a non-negotiable FIT requirement for solo operations and must be briefed explicitly every time.
- Monitor the student's radio calls via the handheld portable radio throughout the pattern portions of the solo. Remain reachable and available for the entire duration of the flight.
- This lesson is closed out by dispatch — ensure the student understands this process and completes their logbook entry independently, including an accurate description of all flight activities.
- Follow up with the student after the solo to debrief any questions or concerns..

Lesson Content (Ground)

Unsupervised solo plan of action

- Normal and crosswind takeoffs
 - Ask the student:
 - What runway are you expecting to take off from?
 - What direction is the wind coming from and how will you apply correction as a result?
- Traffic pattern procedures - Towered
 - Ask the Student
 - What instructions are you expecting after departure?
- Normal and crosswind landings
 - Ask the student:
 - What direction is the wind coming from and how will you apply correction as a result?
- Go-around/rejected landing
 - Ask the student:
 - What situation might cause a go-around?
 - What is the procedure for going around?
 - Perform a go-around if necessary.
 - Unstable approach.
 - Vehicle, wildlife, etc. on the runway.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Solo)**Checklist Procedures**

- Student pilot solo

Preflight Inspection

- Student pilot solo

Operation of the Communications and Navigation Units

- Student pilot solo

Engine Starting

- Student pilot solo

Radio Communication Procedures and ATC Phraseology

- Student pilot solo.

Visual Scanning and Collision Avoidance

- Student pilot solo.

Taxiing

- Student pilot solo.

Normal and Crosswind Takeoff

- Student pilot solo.

Steep Turns

- Student pilot solo.

Maneuvering During Slow Flight

- Student pilot solo.

Stalls

- Student pilot solo.

Ground Reference Maneuvers

- Student pilot solo.

Traffic Patterns

- Student pilot solo.

Go-Arounds or Rejected Landings

- Student pilot solo.

Normal and Crosswind Landings

- Student pilot solo.

Postflight Procedures

- Student pilot solo.

Post-Brief

- Student self-critique.
- Lesson grading.
 - With Primary instructor
- Logbook completion.
 - Description of activities input by student.
- Schedule the next lesson.
- Preview the next lesson → 124.
- Assign homework:
 - Read chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Read chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
 - Read chapter 05 of the Pilot's Handbook of Aeronautical Knowledge - Aerodynamics of Flight
 - Wingtip Vortices.

Completion Standards

This lesson is complete when the applicant can execute a local solo in accordance with all Florida Tech Aviation, FAA and local procedures and regulations.